ME 165
Basic Mechanical Engineering

Lecture 9

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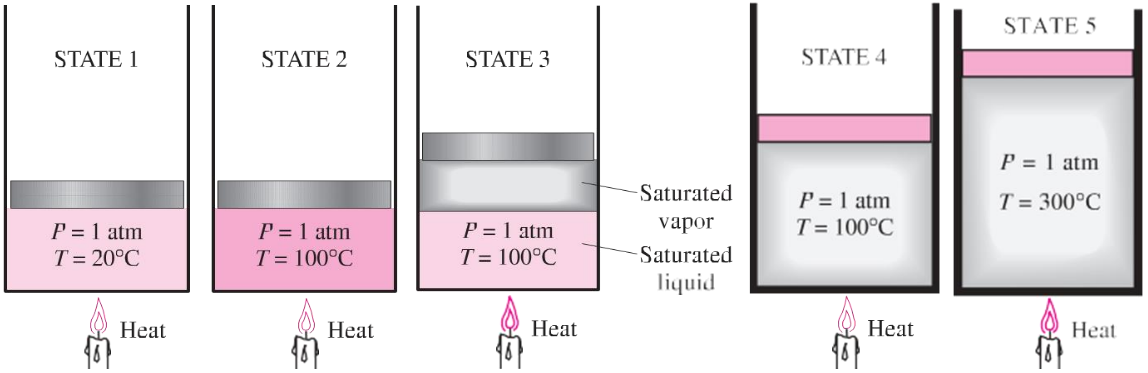
Vapor-Liquid-Solid Phase Equilibrium



1. Compressed ($P > P_{sat}(T)$)

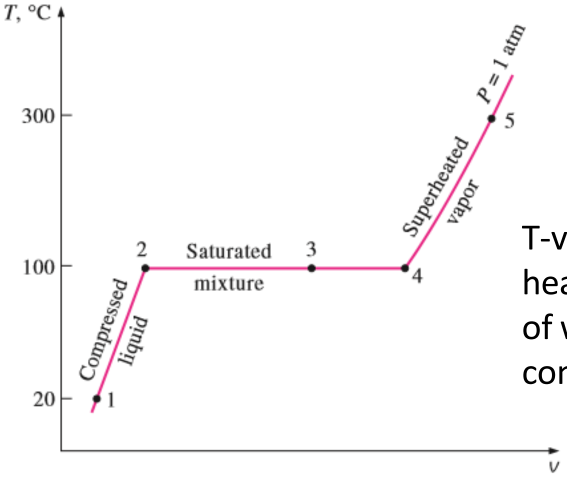
or subcooled liquid ($T < T_{sat}(P)$).

2. Saturated liquid ($T = T_{sat}(P)$).

3. Saturated liquid-vapour mixture ($T = T_{sat}(P)$).

4. Saturated vapour ($T = T_{sat}(P)$).

5. Superheated vapour ($T > T_{sat}(P)$).

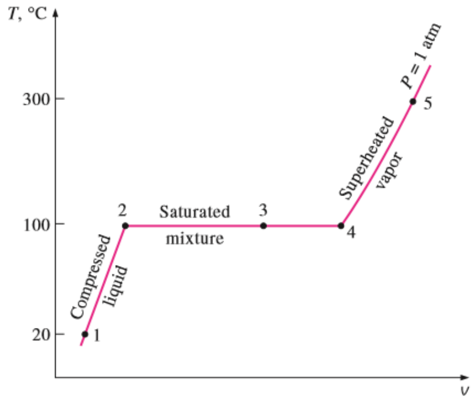


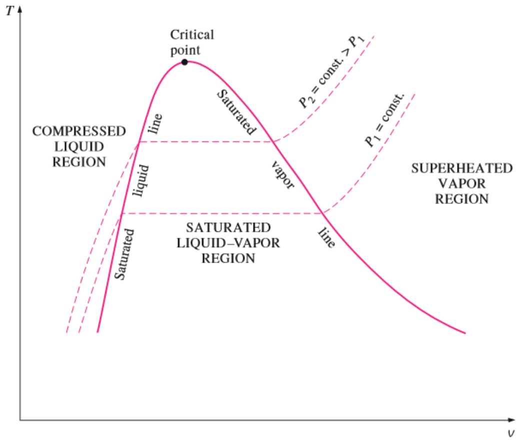
T-v digram for the heating process of water at constant pressure

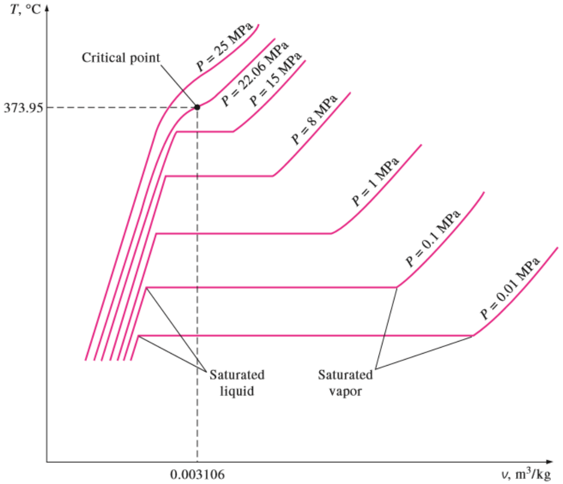
Vapor-Liquid-Solid Phase Equilibrium

- Saturation temperature, T_{sat} is the temperature at which vaporization takes place at a given pressure. This pressure is called the saturation pressure, P_{sat} for the given temperature.
- If a substance exists as liquid at T_{sat} and P_{sat} is called a saturated liquid.
- If the temperature of the liquid is lower than the saturation temperature for the existing pressure, it is called either a sub-cooled liquid (implying $T < T_{sat}(P)$) or a compressed liquid (implying $P > P_{sat}(P)$).
- If a substance exists as vapour at T_{sat}, it is called saturated vapor. When the vapour is at $T > T_{sat}$, it is said to exist as superheated vapor.

T-v Diagram







Boiling point of a liquid **increases** with **increase** in **pressure.** Since a liquid vaporizes when molecules moves faster and farther apart but while **increasing pressure**, you are applying an external force o hold molecules closer to each other.

Throttling Process

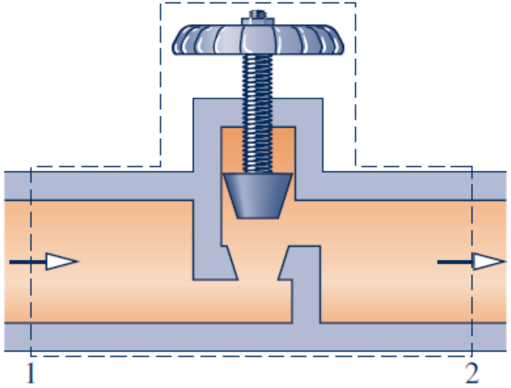
Throttling Process/ Joule-Thomson Effect:

✓When a fluid passes through a restriction such as **a porous plug, a capillary tube, or an ordinary valve**, its pressure decreases.

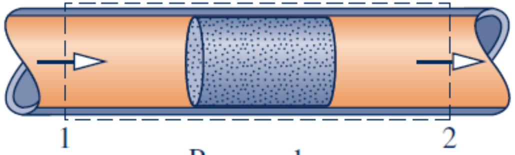
✓The enthalpy of the fluid remains approximately constant during such a throttling process.

✓A fluid may experience **a large drop in its temperature** as a result of throttling, which forms the basis of operation for refrigerators and air conditioners. This is not always the case, however.

✓The temperature of the fluid may remain unchanged, or it may even increase during a throttling process



Partially open valve



Porous plug

Joule-Thomson Effect

- The process in which a gas or liquid at pressure P_1 flows into a region of lower pressure P_2 via a valve or porous plug under steady state conditions and without change in kinetic energy, is called the Joule Thomson process.

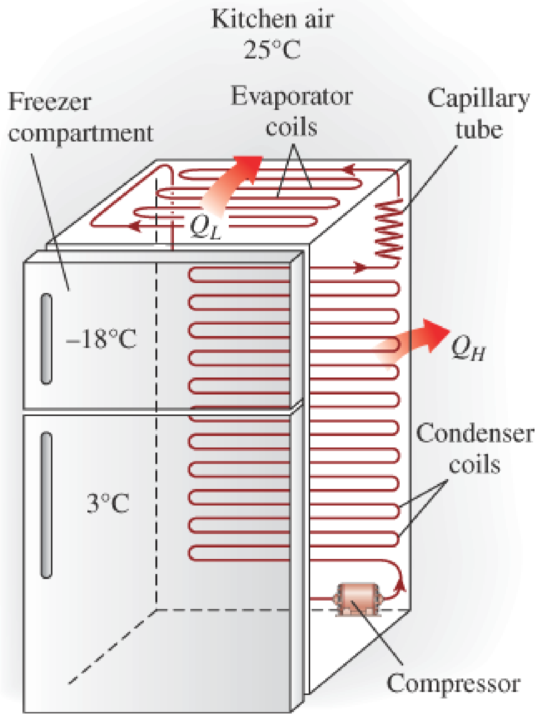
- During this process, enthalpy ($H = U + P*V$) remains unchanged. Typically, as a **fluid expands, the average distance between molecules grows**. Because of intermolecular attractive forces, expansion causes an increase in the potential energy of the gas. If no external work is extracted in the process and no heat is transferred, the total energy of the gas remains the same because of the conservation of energy. The increase in potential energy thus implies a decrease in kinetic energy and therefore in temperature.

- For Ideal gas the temperature remains constant during the throttling process.

Types of Refrigeration Systems

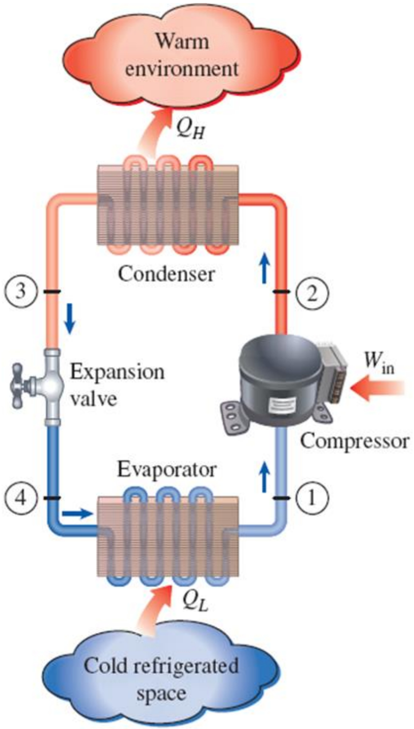
- Vapor Compression Refrigeration System
- Vapor Absorption Refrigeration System
- Steam Jet Refrigeration System
- Thermoelectric Refrigeration

Vapor Compression Refrigeration System



Vapor Compression Refrigeration System

The Vapor Compression Refrigeration system uses a circulating liquid refrigerant as the medium which absorbs and removes heat from the space to be cooled and subsequently rejects that heat elsewhere.



Four Components:

1. **Compressor** (Generally a Reciprocating Compressor)
2. **Condenser** (It condenses the high pressure gaseous refrigerant and acts as storage tank)
3. **Expansion Device** (Also called a throttle valve. May be a globe valve or simply a capillary tube)
4. **Evaporator** (Also termed as cooling coil. It is generally a copper coil with high surface area)

Vapor Compression Refrigeration System

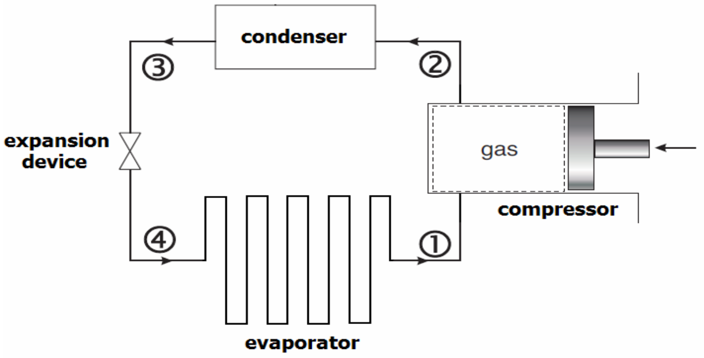
Process:

- **Circulating refrigerant enters** the compressor in the thermodynamic state known as a **saturated vapor** (holding as much water or moisture as can be absorbed) and is compressed to a higher pressure, resulting in a higher temperature as well.
- The hot, compressed vapor is then in the thermodynamic state known as a **superheated vapor** (a vapor that is obtained by raising the temperature of a substance above the saturation temperature while maintaining a constant pressure) and it is at a temperature and pressure at which it can be condensed with typically available atmospheric air.
- That hot vapor is routed through a condenser where it is cooled and condensed into a liquid by flowing through a coil or tubes with relatively cooler atmospheric air flowing across the coil or tubes. This is where the circulating refrigerant rejects heat from the system and the rejected heat is carried away by the atmospheric air.
- The condensed liquid refrigerant, in the thermodynamic state known as a **saturated liquid**, is next routed through an expansion device where it undergoes an abrupt reduction in pressure. That pressure reduction results in the adiabatic flash evaporation of a **part of the liquid** refrigerant. The auto-refrigeration effect of the adiabatic flash evaporation lowers the temperature of the liquid and vapor refrigerant mixture to where it is colder than the temperature of the enclosed space to be refrigerated.

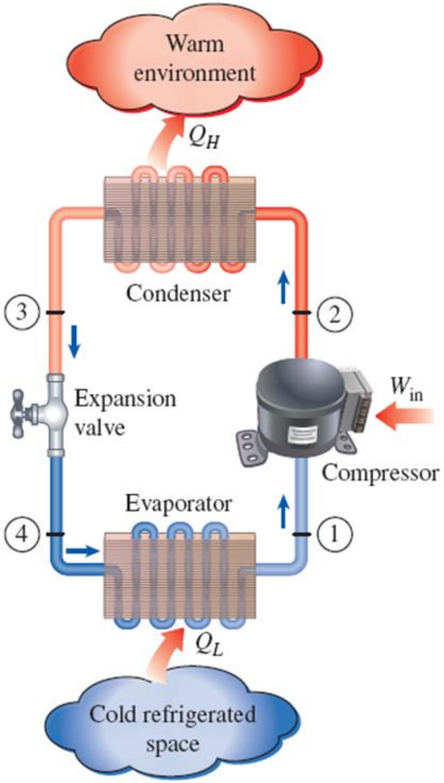
Vapor Compression Refrigeration System

Process:

- The cold mixture is then routed through the coil or tubes in the evaporator. The warm air evaporates the liquid part of the cold refrigerant mixture. At the same time, the circulating air is cooled and thus lowers the temperature of the enclosed space to the desired temperature. The evaporator is where the circulating refrigerant absorbs and removes heat which is subsequently rejected in the condenser and transferred elsewhere by the water or air used in the condenser.
- To complete the refrigeration cycle, the refrigerant vapor from the evaporator is again a saturated vapor and is routed back into the compressor



Vapor Compression Refrigeration System



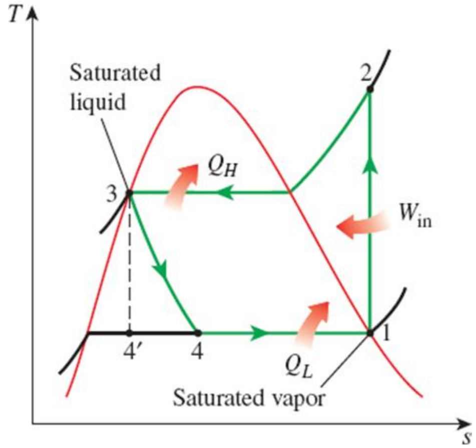
Ideal Vapor Compression Refrigeration Cycle

1—2 : Isentropic compression in a compressor

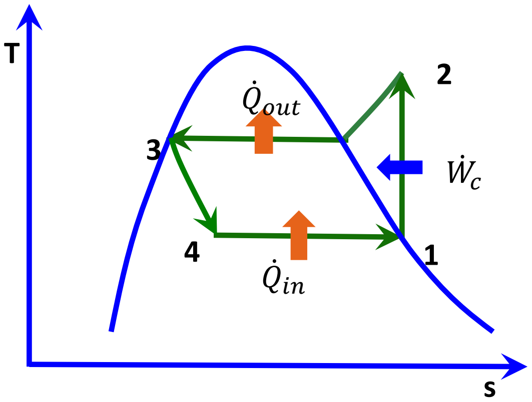
2—3 : Constant Pressure heat rejection in a condenser

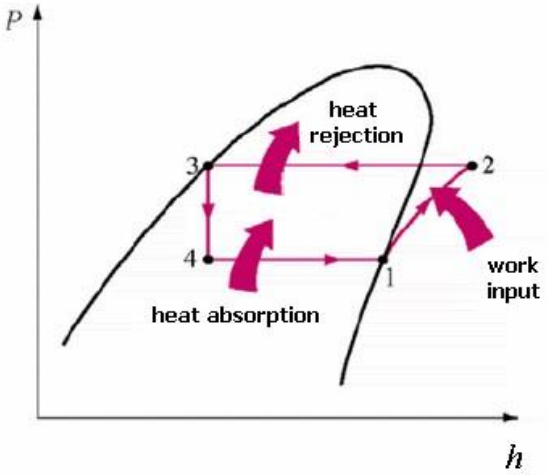
3—4 : Throttling in an expansion device (Enthalpy constant)

4—1 : Constant pressure heat absorption in an evaporator



Vapor Compression Refrigeration System





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Engineering model (Assumptions):

- Each component is analyzed as a control volume at steady state.
- Dry compression is presumed.
- Kinetic and potential energy changes are ignored.

Cooling effect (Desired output), $\dot{Q}_{in} = \dot{m}\,(h_1 - h_4)$

Work input (Required input), $\dot{W}_c = \dot{m}\,(h_2 - h_1)$

$$\text{COP} = \frac{\text{Desired output}}{\text{Required input}} = \frac{\dot{Q}_{in}}{\dot{W}_c} = \frac{h_1 - h_4}{h_2 - h_1}$$

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